

30/09/2026

let's find the potentials f_1, f_2, f_3

s.t.

$\text{grad } f_1 = F$ in Ω_1 , $\text{grad } f_2 = F$ in Ω_2 , $\text{grad } f_3 = F$ in Ω_3 .

Find $f_1 \in C^2(\Omega_1)$

$$\left(\frac{\partial f_1}{\partial x}, \frac{\partial f_1}{\partial y} \right) = (F_1, F_2) = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2} \right)$$

$$\bullet \quad \frac{\partial f_1}{\partial x} = \frac{-y}{x^2+y^2} \rightarrow \int \frac{\partial f_1}{\partial x} dx = \int \frac{-y}{x^2+y^2} dx \rightarrow f_1 = -\arctan\left(\frac{x}{y}\right) + \alpha(y)$$

$$\bullet \quad \frac{\partial f_1}{\partial y} = \frac{x}{x^2+y^2} \quad \cdot \quad \frac{\partial}{\partial y} \left(-\arctan\left(\frac{x}{y}\right) + \alpha(y) \right) =$$

$$= \frac{\cancel{x}}{\cancel{x^2+y^2}} + \alpha'(y) = \boxed{\frac{\cancel{x}}{\cancel{x^2+y^2}}}$$

$$\alpha'(y) = 0. \rightarrow \int \alpha'(y) dy = \int 0 dy \rightarrow \alpha(y) = c_1, \quad c_1 \in \mathbb{R}$$

$$\boxed{f_1 = -\arctan\left(\frac{x}{y}\right) + c_1}$$

- find $f_2 \in C^2(\Omega_2)$ s.t. $\text{grad } f_2 = F$ in Ω_2
 $\rightarrow$ the same as before $\rightarrow f_2 = -\arctan\left(\frac{x}{y}\right) + c_2, \quad c_2 \in \mathbb{R}.$

- find $f_3 \in C^2(\Omega_3)$ s.t. $\text{grad } f_3 = F$ in Ω_3

$$f_3 = \begin{cases} -\arctan\left(\frac{x}{y}\right) + c_1 & \text{if } y > 0 \\ -\arctan\left(\frac{x}{y}\right) + c_2 & \text{if } y < 0 \\ \text{?} & \text{if } y = 0 \text{ and } x > 0 \end{cases}$$

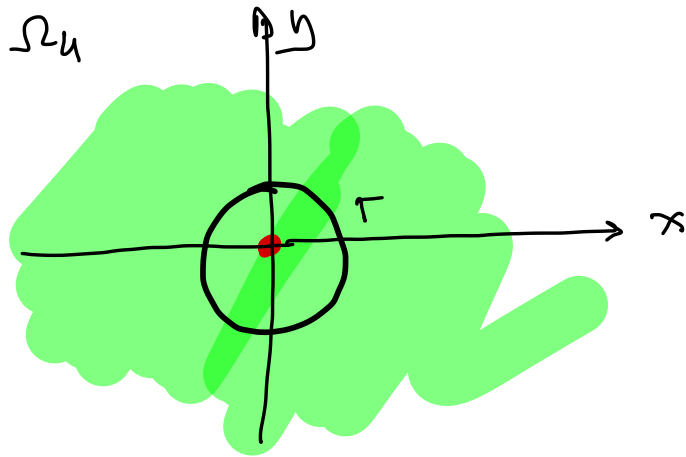
let's enforce f_3 to be continuous on $y=0$ and $x>0$

$$\lim_{\substack{y \rightarrow 0^+ \\ x > 0}} f_3 = \lim_{\substack{y \rightarrow 0^+ \\ x > 0}} \left(-\arctan\left(\frac{x}{y}\right) + c_1 \right) = -\frac{\pi}{2} + c_1$$

$$\lim_{\substack{y \rightarrow 0^- \\ x > 0}} f_3 = \lim_{\substack{y \rightarrow 0^- \\ x > 0}} \left(-\arctan\left(\frac{x}{y}\right) + c_2 \right) = +\frac{\pi}{2} + c_2$$

$$-\frac{\pi}{2} + c_1 = \frac{\pi}{2} + c_2 \rightarrow c_1 = \pi + c_2$$

$$f_3 = \begin{cases} -\arctan\left(\frac{x}{y}\right) + \overbrace{\pi}^{c_1} + c_2 & \text{if } y > 0 \\ -\arctan\left(\frac{x}{y}\right) + c_2 & \text{if } y < 0 \\ \frac{\pi}{2} + c_2 & \text{if } y=0 \text{ and } x > 0 \end{cases}$$



$$\Gamma = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1 \}$$

$$\gamma: [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto \gamma(t) = (\cos t, \sin t)$$

$$\gamma'(t) = (-\sin t, \cos t)$$

$$\int_{\Gamma} F \cdot d\mathbf{l} \neq 0 \quad ?$$

$$\int_{\Gamma} F \cdot d\mathbf{l} = \int_0^{2\pi} F(\gamma(t)) \cdot \gamma'(t) dt$$

$$= \int_0^{2\pi} \left(\frac{-\sin t}{\underbrace{\sin^2 t + \cos^2 t}_{=1}}, \frac{\cos t}{\underbrace{\sin^2 t + \cos^2 t}_{=1}} \right) \cdot (-\sin t, \cos t) dt$$

$$= \int_0^{2\pi} \sin^2 t + a^2 t \, dt = \int_0^{2\pi} dt = 2\pi \neq 0 \Rightarrow F \text{ does not derive from a potential in } \Omega_4$$

2.4 Green's theorem (all the results in this section are for $n=2$)

2.4.1 Recall, notations, and definitions

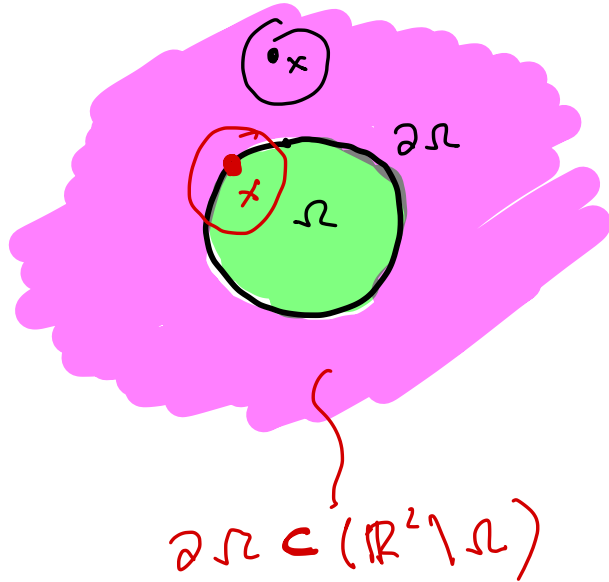
- $D_r(x) = \{ y \in \mathbb{R}^2 : \|x - y\|^2 < r^2 \}$

on open disk centered at x , with radius r



- we denote as $\partial\Omega$ the boundary of Ω

$$\partial\Omega = \{ x \in \mathbb{R}^2 : \Omega \cap D_r(x) \neq \emptyset \text{ and } (\mathbb{R}^2 \setminus \Omega) \cap D_r(x) \neq \emptyset, \forall r > 0 \}$$

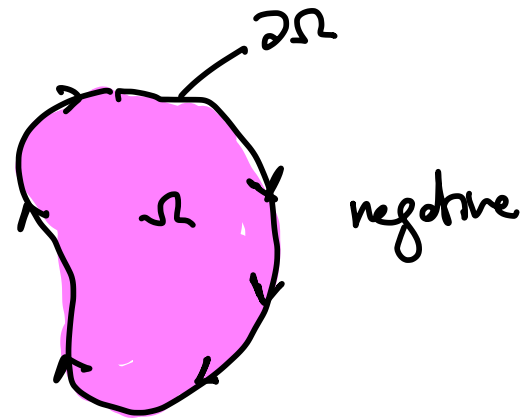
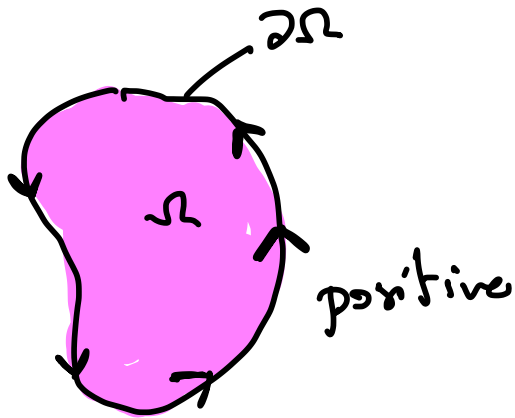


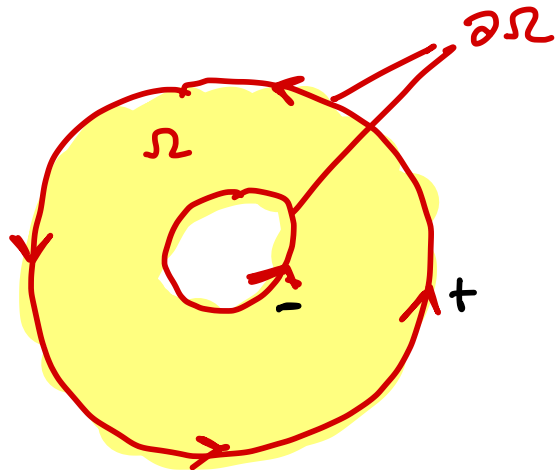
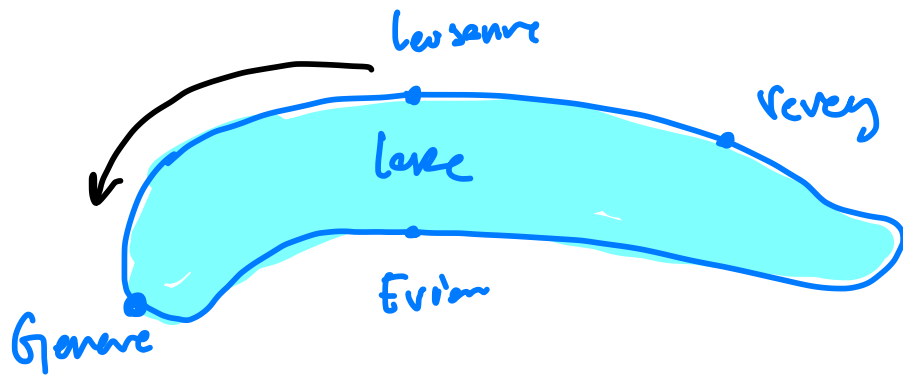
• we denote $\bar{\Omega} = \Omega \cup \partial\Omega$
as the closure of Ω :

the set of points of $\mathbb{R}^2$ that
do not belong to the interior of
 $\mathbb{R}^2 \setminus \Omega$

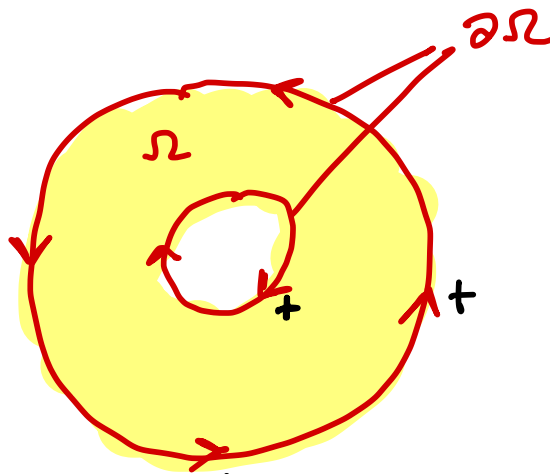
Definitions: *(collection)*

1) Let $\Omega \subset \mathbb{R}^2$ be an open bounded domain s.t. $\partial\Omega$ is a closed (piecewise) simple regular curve. We say that the parameterizations of $\partial\Omega$ are positively (or negatively) oriented if when we circulate every single curve the domain is on the left (or, resp, on the right).



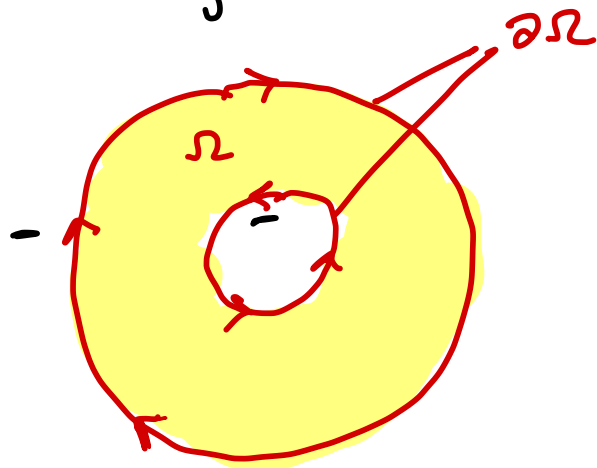


not well oriented



positive.

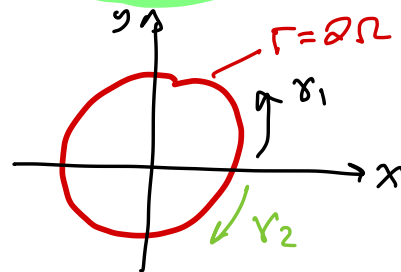
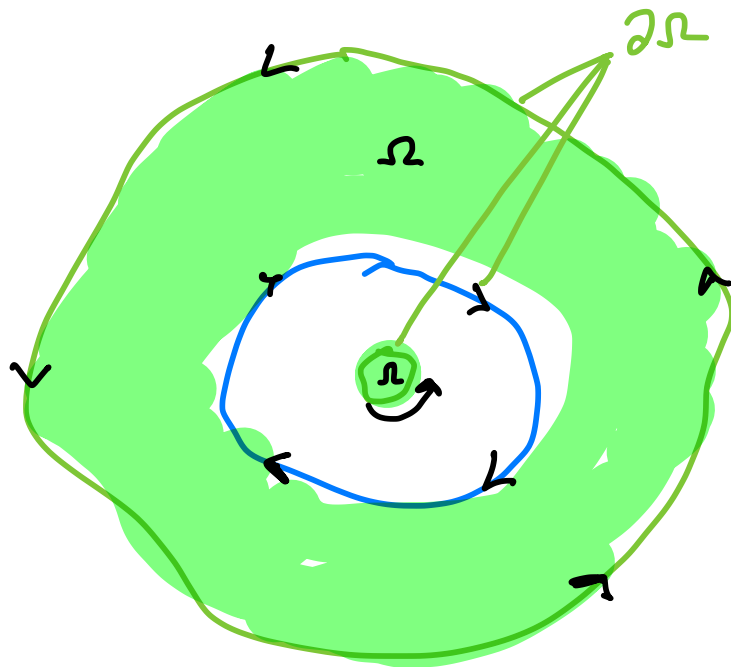
negative



Example

$$\Omega = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1 \}$$

$$\partial\Omega = \Gamma = \{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1 \}$$



$$\gamma_1: [0, 2\pi] \rightarrow \mathbb{R}^2$$

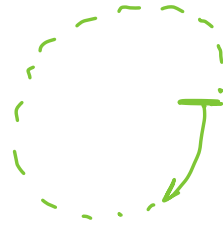
$$t \mapsto (\cos t, \sin t)$$



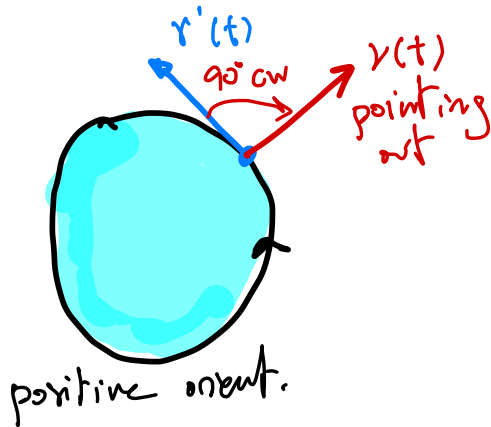
$$\gamma_2: [0, 2\pi] \rightarrow \mathbb{R}^2$$

$$t \mapsto (\cos(-t), \sin(-t))$$

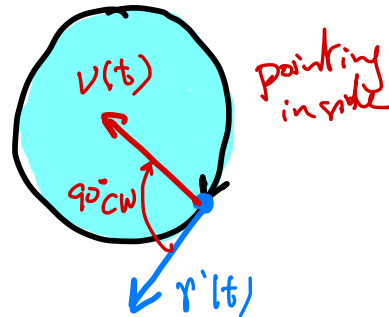
$$= (\cos t, -\sin t)$$

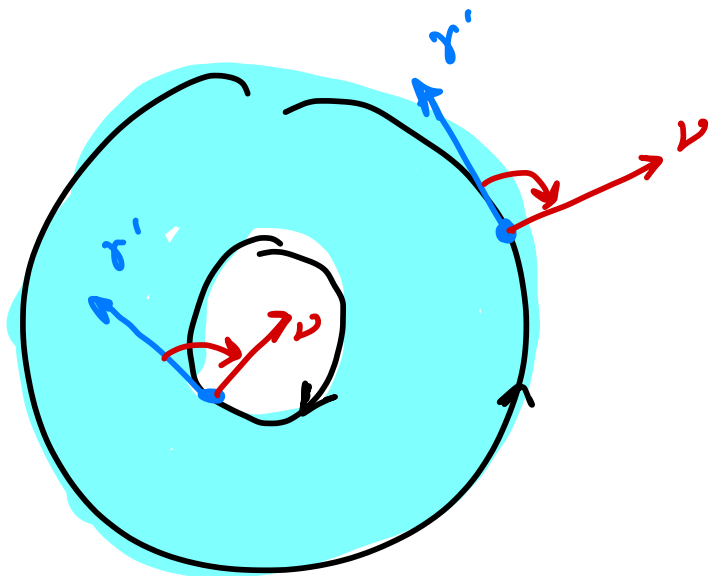


Convention: to compute the normal vector to a curve in a point $\gamma(t)$ we rotate the tangent vector $\gamma'(t)$ 90° in clockwise direction.

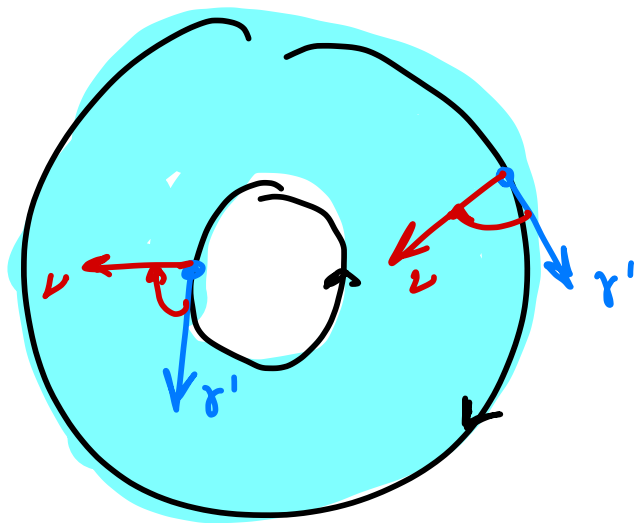


negative orient.





positive
normals point out



negative
normal point inside.

2) we say that a bounded open domain $A \subset \mathbb{R}^2$ is
a regular domain if $\exists$ bounded open domains $A_0, A_1, A_2, \dots, A_m \subset \mathbb{R}^2$

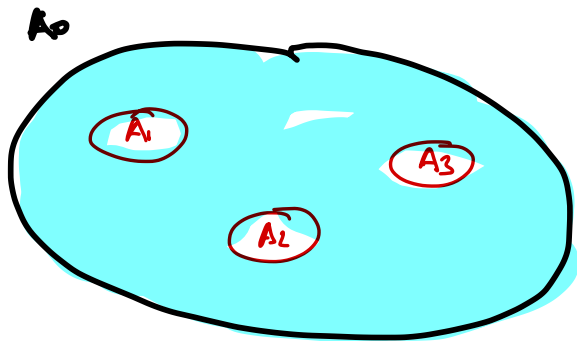
s.t.:

- $A = A_0 \cup \bigcup_{i=1}^m \bar{A}_i$

- $\bar{A}_j \subset A_0, \forall j=1, 2, \dots, m$

- $\bar{A}_i \cap \bar{A}_j = \emptyset, \forall i, j=1, 2, \dots, m, i \neq j$

- $\partial A = \cup (\partial A_j), j=0, 1, \dots, m.$



2.4.2 Green's theorem

Theorem:

let $A \subset \mathbb{R}^2$ be a regular domain whose boundary ∂A is positively oriented. let $F: \bar{A} \rightarrow \mathbb{R}^2$

$$(x, y) \mapsto F(x, y) = (F_1(x, y), F_2(x, y))$$

be a vector field s.t. $F \in C^1(\bar{A}, \mathbb{R}^2)$. Then

$$\iint_A \text{curl } F(x,y) \, dx \, dy = \iint_A \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \, dx \, dy = \int_{\partial A} F \cdot d\mathbf{h}$$

Remark: If F derives from a potential $\Rightarrow \text{curl } F = 0 \Rightarrow$ ↑ Thm 1.

$$\Rightarrow \iint_A \text{curl } F \, dx \, dy = 0 = \int_{\Gamma} F \cdot dl$$

→ relates to Thm 2.